UNIVERSITY OF LIVERPOOL | English Language Centre

COMP201: Academic Language and Skills

Session 1: Use-Case Diagrams and Descriptions

Session aim

The main aim of this session is to help you understand and communicate use-case diagrams clearly for your assignment. We will:

- Analyse a library system to identify human actors and use-cases
- Create a use-case diagram for the system
    - Evaluate a student example and compare it with yours
    - Watch an example solution from a teacher
- Identify human actors and use-cases in your assignment system
- Review an example use-case description to understand how to structure and write them effectively

Task 1: Analysing an example system

Why: To identify key actors and interactions in a system before making a use-case diagram.

We are going to use an example of **a book borrowing system in a library** to help understand use-cases in preparation for your assignment.

1. **In pairs or small groups, discuss the following:**
    1. Who are the **two** main users ('human actors') of a book borrowing system in a library?
    2. How do those human actors interact with the book borrowing system? What are some of the common tasks they need to do?
    3. Make a list of possible use-cases involving the human actors. Discuss what will happen in each of those cases.

2. **Read the description of a system below and identify all of the use cases involving human actors.**

COMP 201 - Use Case Diagram Task

Draw a use case diagram for the system below. Consider only human actors.

Read the following requirements about a Library:
The library requires a new borrowing system, which keeps track of librarians, members, books, and borrowing records. The information held about librarians will be: name, staff number, and area of expertise. The information held about members will be: name, address, member number, current loans, previous loans, current late fee, and previous payments. The information held about books will be: title, author, ISBN, Dewey decimal classification number, and status (available, or on loan). The system will allow the librarian to query any system data, add new books, delete old books, add new members, delete old members, issue books to members, return books from members, and to collect late fees from members. Any system use that edits a member's record requires authentication by means of membership card or proof of address (e.g. a recent utility bill). The system will allow a member to issue a book, return a book, query personal account information, and to pay a late fee. This will be done using a self-service kiosk, after authenticating using their library card.

3. **VIDEO: Check your answers by watching the first part of the video** (from 00.00 to 01.47)

Suggested answers below

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COMP 201 - Use Case Diagram Task

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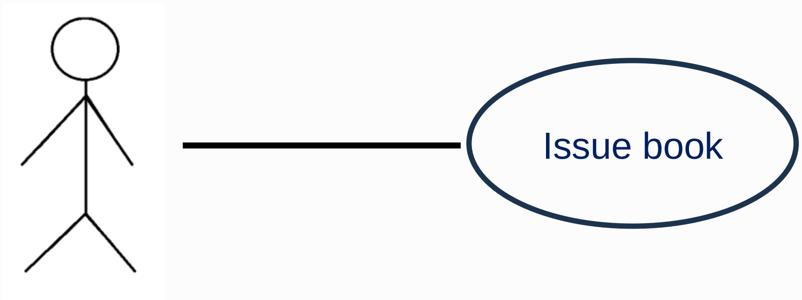
Task 2: Drawing a use-case diagram

Why: To practice drawing use-case diagrams to organise and present system interactions.

You are going to practise drawing a use-case diagram. You will then compare your diagram with a student example, and finally we will see the lecturer's suggestion.

1. **Work in pairs or small groups. Draw a use-case diagram based on the library scenario. Follow the guidance below.**
    - Make a hand-drawn diagram, or use whatever software is easiest (e.g. app.genmymodel.com).
    - Be prepared to describe how you have organised your diagram.
    - The diagram does not need to be perfect – it just needs to convey the key information.
    - You can represent each human actor with a simple drawing, and each use-case in an oval shape, like the examples below. Connect the use-cases to the relevant human actor with a line.



Above: (left) example representation of a human actor; (right) example use case

2. **Share your diagram with another group of students. Explain how you have designed it.**

3. **Your teacher will show you a student's use-case diagram. Evaluate the diagram. Is it similar or different to yours? (How?)**

4. **VIDEO: Watch rest of the video example and compare the teacher's use-case diagram with your own.**

Task 3: Analysing the assignment system description

Why: To analyse the assignment system and identify relevant actors and use-cases.

Before you begin creating your use-case diagram and writing your use-case descriptions, you will need to read the description of the proposed system, then identify all the **human actors** and also the important **use-cases**.

Assignment Tip: Human actors

The assignment instructions tell you which human actors you need to consider: customer, operator and engineer.

1. **Read the system description quickly and identify which sections of the text relate to each of the three human actors.**

Proposed drinks machine

Proposed drinks machine control system

Your company has been commissioned to design a drinks vending machine which will be able to produce a range of hot beverages, for example

Hot chocolate, standard coffee, flat white, cappuccino, tea and soup. For tea and standard coffee only the user can pick with milk and/or sugar.

All of these drinks are made by heating up water, then mixing it up in a mixing vessel, the mixture is then dispensed into a disposable cup, the user also has an option to use their own cup. After each drink is dispensed the mixing vessel is cleaned out by running hot water through it for a short amount of time.

To do all this, the machine needs to be able to:

Accept and validate coins and calculate the current value of the coins inserted

Accept instructions to choose a particular drink recipe

Heat up the water to the correct temperature then mix the drinks

Dispenses and fill the disposable cups for the drinks

The system has the following sensors, 1 cup out of stock sensor, 16 drink ingredients out of stock sensors, cash box full sensor, water temperature sensor, door open sensor.

The machine has a keypad on the front with a keypad as shown in Figure 1, this is used to make the drink selection and to control the engineer's tests. The machine also has an alphanumeric LED display.

Figure 1 Keypad and alpha display

The drinks machine is fitted with a GPRS/3G/4G/5G WiFi card which allows it to be connected to the internet and be controlled by a service operator. The service operator can find out the current stock and cash levels in the machine as well as be informed via an alert if the machine's stock levels are low or the cash box is full. The operator can also make requests to download accounts information from the machine for a particular accounting period, for example 4/4/2021 to 4/5/2021. Operator access to the machine is protected using a username and password stored within the machine. There is also a master password available which can be used create new accounts in case the machines memory has been wiped due to hardware failure. Also the recipes for the drinks can be downloaded to the machine online.

When the machine requires service (such as needing more coffee or the cash to be collected or has run out of cups), a service engineer visits the machine. The back of the machine is opened, this puts the machine into service mode, the engineer will then collect the cash and does a series of tests by using the keypad. Each engineer has a unique passcode which they need to type into the keypad after opening the service door. If they don't do this within a short time interval the machine goes into alarm mode and sends an alert to the service operator.

When the cash is collected, the engineering must record the amount they collect into the keypad, this data is stored in the machine and can be retrieved by the service operator.

Keyboard coding scheme Given that the engineer and customer both have to use the keypad, for your solution please devise a coding for at least all the drink codes. For this part of the solution, try to make it so the customers only need to press the minimal number of key presses possible to get their drink and also make it impossible to get the wrong drink if they only get 1 key press incorrect (i.e. 1 key press wrong should give an invalid key press sequence). This type of code would be said to have a hamming distance of 2 key presses. Note the keyboard only has the following characters 0,1,2,3,4,5,6,7,8,9,A,B,C,D, E and F.

Scheme should be presented as follows (note this example is not correct)

Drink	Code
Coffee	1
Tea	2

etc

Assignment Tip: Use cases

To help you identify use-cases, you can look for verb + noun collocations (a collocation is when two words frequently occur together).

Examples from the library system we saw earlier:

Verb + noun phrase

return + a book

issue + a book

pay + a late fee

2. **Read the 'proposed drinks machine' system text again. Identify as many verb-noun collocations as you can. Try to find some for each of the human actors.**
   (You don't need to find them all but it's good to have options to choose from later)

Important:

- **Please note that the examples highlighted in the text above are to show verb + noun collocations that could be a use-case.**
- **I am not saying that these are use-cases you need to include in your assignment.**

Task 4: Writing use-case descriptions

Why: To understand how to structure and write clear use-case descriptions using an example.

Assignment Tip: What's next?

✓ You have identified the **human actors** and the important **use-cases** in the proposed system.

➢ Now you need to **select 15 important use-cases**, including some for each human actor.

➢ When you have selected 15 use-cases, you need to **write use-case descriptions** for each of them, using the template from the assignment instructions.

We are going to look at an example from the library book lending system to see how use-case descriptions are written.

1. **Read the use-case description table for "Issue Book to Member (Librarian)" carefully. Pay attention to each section. Discuss the following questions:**
   1. What does the Pre-conditions section tell you?
   2. Why are the steps in the Event flow section numbered?
   3. What happens in the Post-condition section?

ID	UC1
Actors	Librarian, Member
Name	Issue Book to Member
Description	The librarian issues a book to a member, updating the system to reflect the book’s status as on loan.
Pre-conditions	The member is authenticated (using a library card or proof of address). The book is available in the system.
Event flow	1. The librarian verifies the member’s identity using a library card. 2. The librarian checks the availability of the requested book in the system. 3. The librarian assigns the book to the member’s account and marks it as "on loan". 4. The system updates the borrowing record for both the member and the book.
Post-condition	The book’s status is updated to "on loan", and the member’s borrowing record is updated.
Includes	Authenticate member
Extensions	None
Triggers	Member requests to borrow a book.

Suggested answers below

1. **Read the use-case description table for "Issue Book to Member (Librarian)" carefully. Pay attention to each section. Discuss the following questions:**
    1. What does the Pre-conditions section tell you?
        - It lists the conditions that must be true before the use case can begin.
        - In this case, it ensures that the member is authenticated (using a library card or proof of address) and that the book is available in the system.
        - It tells us what needs to be in place for the use case to proceed.

    2. Why are the steps in the Event flow section numbered?
        - To show the specific sequence of actions that occur during the use case.
        - Numbering helps organize the flow of events and ensures clarity, so the reader understands the correct order of operations.
        - It also makes it easier to follow and implement the process step by step.

    3. What happens in the Post-condition section?
        - It describes the state of the system after the use case is completed.
        - In this case, it explains that the book's status is updated to "on loan," and the member's borrowing record is updated.
        - It confirms the outcome or effect of the use case on the system.

In the next session

Session 2 - We will focus on Task 2 ('non-functional' requirements)

- Analyse the assignment instructions for this section
- Clarifying 'non-functional' requirements
- Read some example descriptions
- Identify language features typical of these descriptions
- Begin planning and writing your description